

Discussion Section 1

GOVT 6039

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What quantity would answer this question?

Does making local news free
make people better informed?

?

Write down one quantity you would report if you could observe the entire population. Words are enough. A formula is better.

Three targets behind one sentence

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Y score on a local-politics quiz, from 0 to 20

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average knowledge

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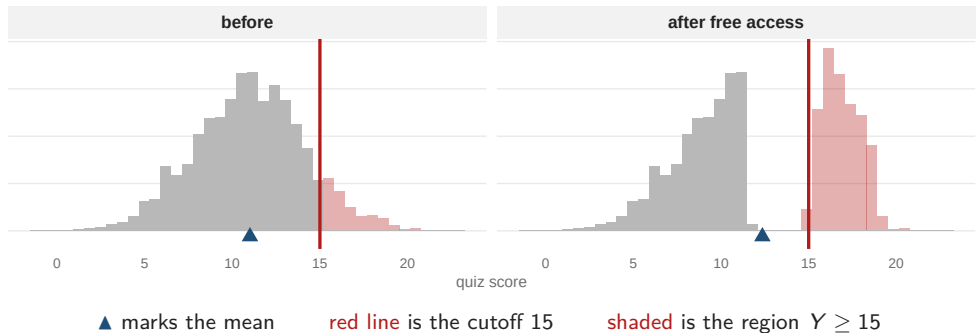
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The data do not choose among these. The research question must.

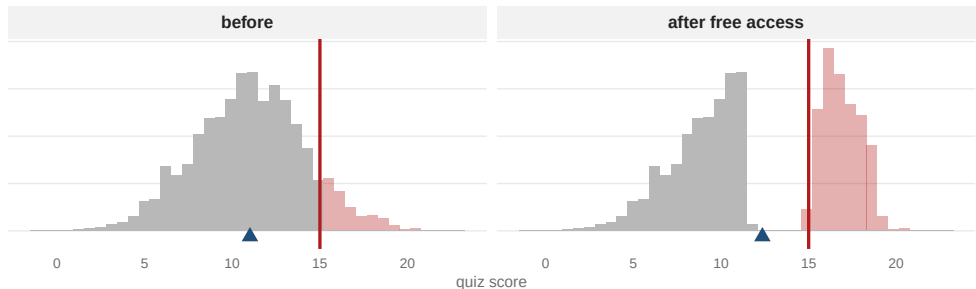
Same intervention, different summaries

Suppose free access raises the scores of people who were already close to the cutoff, and leaves everyone else where they were.



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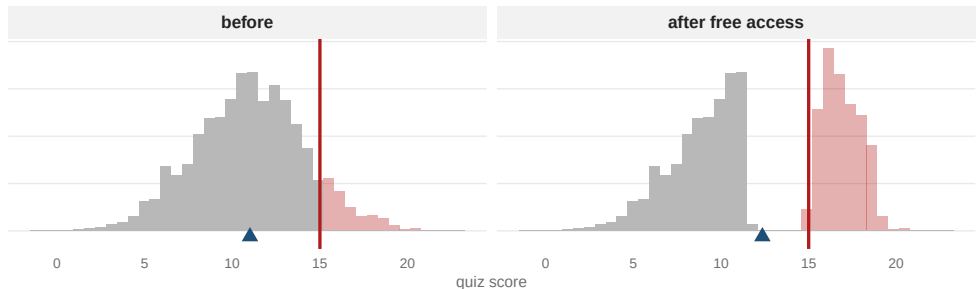
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shaded is the region $Y \geq 15$

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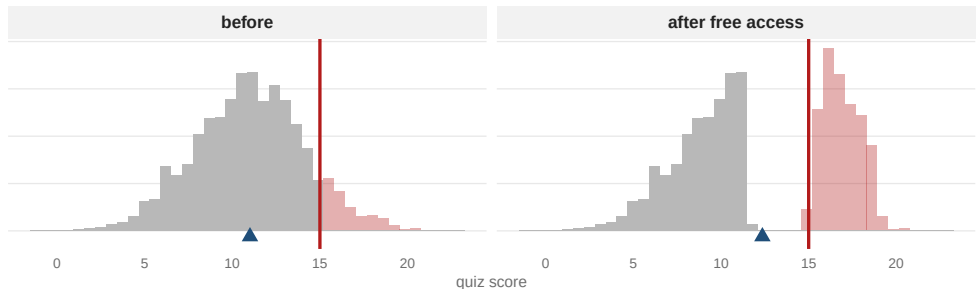
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Same intervention; different summaries of what changed.

Target, rule, or result?

H is hours per week a person spends with local news. D is the dataset we happen to have.

- 1 **estimand** — a feature of the population, fixed before any data
- 2 **estimator** — a procedure, a rule you could run on any dataset
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$f(6)$ is **1** · the cubic procedure is **2** · 14.1 is **3**

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?

We draw a second sample from the same population and run the same procedure. Which of the four objects take different values?

The pipeline



$P \longrightarrow D$ the population also generates the dataset

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D and $\hat{\theta}$ take new values. θ does not move. $\mathcal{A}$ moves only if the analyst rewrites it.

Which edge broke, and what would repair it?

Three studies of the same question: does free access to local news raise political knowledge?

- A** Two independent samples of 300 people give free-access gaps of 2.1 and 0.4 points.
- B** The target is the gap among people who already follow local politics. The fitted model has no interaction term, so it reports one gap for followers and non-followers alike.
- C** The policy question is whether people cross a threshold of being informed. The study reports the mean quiz score and nothing else.

?

For each case, name the edge that broke and one repair.

Different breaks need different repairs

Failure

What went wrong

Appropriate response

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C is the first row, B the second, A the third.

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C is the first row, B the second, A the third.

More data addresses sampling uncertainty. It does not redefine the target, nor enlarge what the model can represent.

Four cells, three questions

A = 1 if offered free access. Randomly assigned.

Z = 1 if the person already follows local politics closely. Observed, not assigned.

	$A = 0$	$A = 1$
$Z = 1$	18	19
$Z = 0$	10	15

each cell is $E[Y \mid A, Z]$

Y = quiz score. The four numbers below are population means, known exactly.

π = $P(Z=1) = 0.5$, the share of close followers.

- 1 **Level.** Expected knowledge among people offered access *and* already following.
- 2 **Heterogeneity.** How much smaller is the access gap among followers than among non-followers?
- 3 **Population average.** The access gap for one person drawn at random.

? For each: a number, an expression in the four cells, and one sentence saying what it means.

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$$\theta_L = E[Y \mid A=1, Z=1] = 19$$

one cell, read off directly

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	$A = 0$		$A = 1$
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$Z = 0$	10	$\Delta_0 = 5$	15

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$\Delta_1 = 19 - 18 = 1$ and $\Delta_0 = 15 - 10 = 5$ the gap inside each row

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$$\theta_H = \Delta_1 - \Delta_0 = 1 - 5 = -4 \quad \text{how much smaller the gap is for followers}$$

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	$A = 0$	$A = 1$	
$Z = 1$	18	19	$\times \pi$
	$\Delta_1 = 1$		
$Z = 0$	10	15	$\times (1 - \pi)$
	$\Delta_0 = 5$		

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$$\theta_A = (1 - \pi)\Delta_0 + \pi\Delta_1 = 0.5(5) + 0.5(1) = 3 \quad \text{each gap weighted by how common its row is}$$

The same four cells, written as coefficients

$$f(a, z) = E[Y \mid A = a, Z = z] = \underbrace{\beta_0}_{\text{cell } (0,0)} + \underbrace{\beta_1 a}_{\text{shift from } A} + \underbrace{\beta_2 z}_{\text{shift from } Z} + \underbrace{\beta_3 az}_{\text{extra shift when both}}$$

$$\beta_0 = 10 \quad \beta_1 = 5 \quad \beta_2 = 8 \quad \beta_3 = -4$$

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$$\theta_H = \beta_3$$

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θ_A averages the two *gaps*, Δ_0 and Δ_1 . It does not average θ_H .

The short model reports 3. Which target is that?

The model has no interaction term, so one A -gap is forced onto both rows.

$$\hat{f}_{\text{short}}(a, z) = \hat{\alpha}_0 + \hat{\alpha}_1 a + \hat{\alpha}_2 z$$

$$\hat{\alpha}_1 = 3$$



one gap, forced onto both rows

? Which of the three targets does the number 3 answer?

- A** the gap among non-followers, $\Delta_0 = 5$
- B** the gap among followers, $\Delta_1 = 1$
- C** the population-average gap, $\theta_A = 3$
- D** all three at once

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A restricted model can be exactly right about an average while being incapable of representing either subgroup relationship.

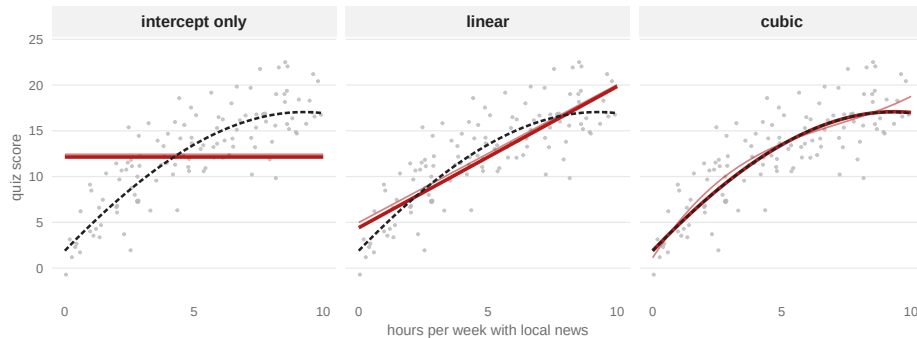
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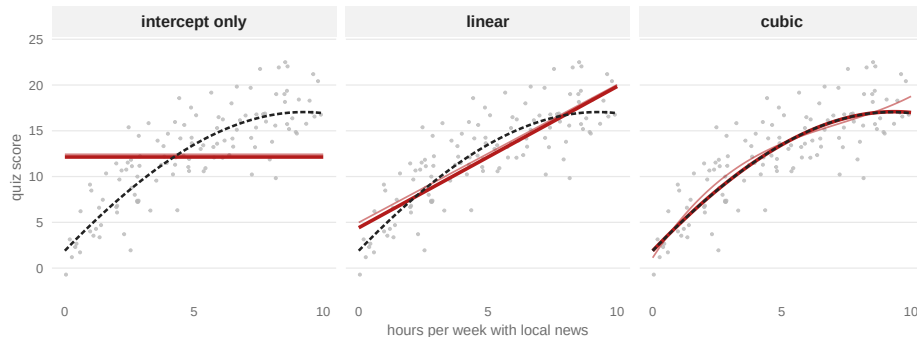
Controlling for Z is not the same as letting the A -gap vary with Z .

Infinite data cannot escape the model class



dotted black: the truth f · solid red: best member of the class · thin red: one finite sample

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Which of the two gaps survives when the sample becomes infinite?

Two different gaps

$$\underbrace{f_{\mathcal{F}}^*}_{\text{best member of the class}} = \arg \min_{g \in \mathcal{F}} \underbrace{E[(f(H) - g(H))^2]}_{\text{average squared distance to the truth, over the population}}$$

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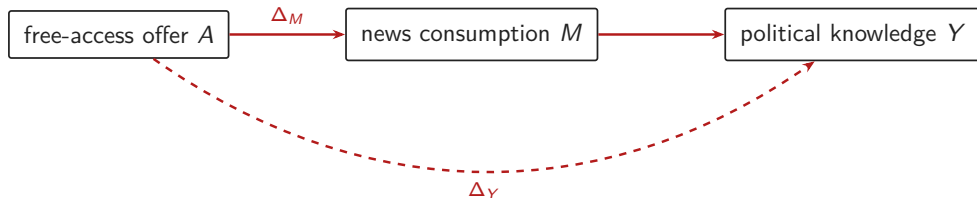
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Least squares finds the best fit to the sample. The solid red curve is the best member in the population, and no single dataset shows it to you.

Offering access can change knowledge only through reading

In Trexler's experiment the researcher controls the offer, not the reading. Everything downstream is a consequence.

randomly assigned

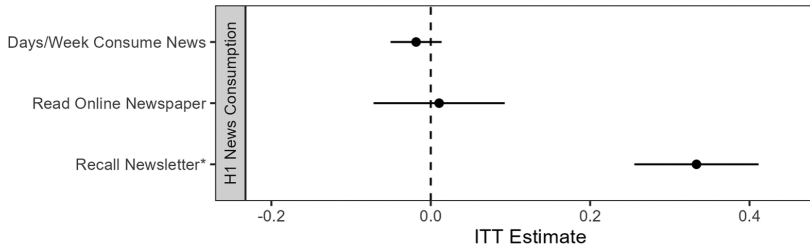


$$\Delta_M = \underbrace{E[M \mid A=1] - E[M \mid A=0]}_{\text{did the offer change reading?}}$$

$$\Delta_Y = \underbrace{E[Y \mid A=1] - E[Y \mid A=0]}_{\text{did the offer change knowledge?}}$$

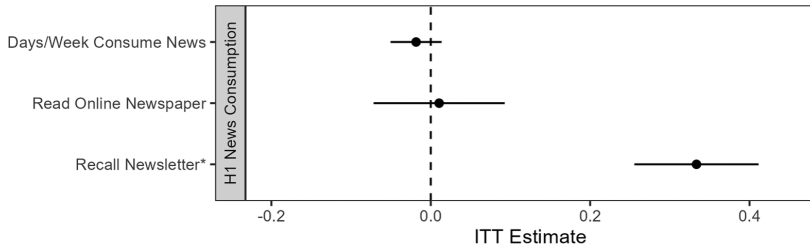
Both compare people by the offer received, not by what they did with it: intent-to-treat quantities.

Did the offer change reading?



Adapted from Trexler (2026), *JOP* 88(3), figure 2. Each row is one outcome. Dot is the ITT estimate, bar is the 95% interval, dashed line is zero.

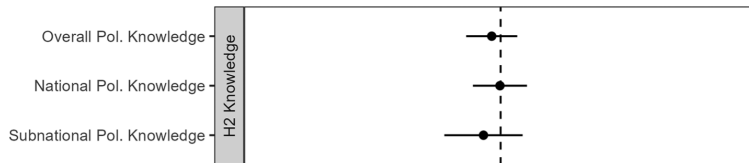
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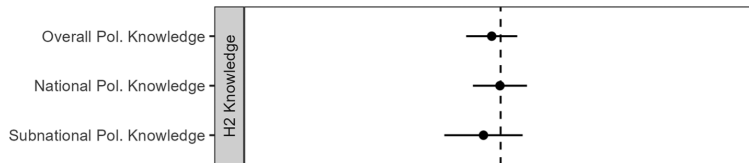
Newsletter recall goes up by about 0.33. The two measures of actual consumption sit on zero.

Did it change knowledge?



Adapted from Trexler (2026), *JOP* 88(3), figure 3, knowledge block. Horizontal axis runs from -0.2 to 0.2 .

Did it change knowledge?



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All three knowledge estimates overlap zero.

“Reading local news does not make people better informed.”

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Rewrite this so the treatment, the outcome, the population and the design all match what was measured.

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Offering two months of free digital access raised newsletter recall but did not measurably raise reported news consumption or political knowledge among the registered non-subscribers who joined the study.

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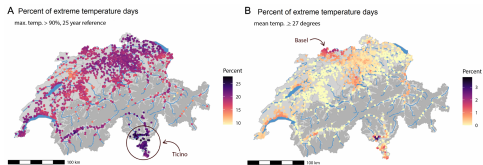
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The experiment estimates the effect of an offer, not the effect of reading.

The same weather history, two different treatments

Quoß and Rudolph ask whether extreme weather shifts climate policy preferences. Someone must first decide what counts as extreme.



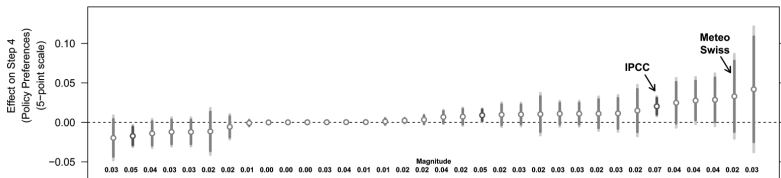
Adapted from Quoß & Rudolph (2026), *JOP* 88(3), figure 4. Each dot is one respondent. A: IPCC definition. B: MeteoSwiss definition.

$$\underbrace{W_i^{(\text{IPCC})} \neq W_i^{(\text{MeteoSwiss})}}_{\text{the same person gets a different exposure value}} \quad \theta^{(k)} = T(P; W^{(k)})$$

? Two rival estimators of one estimand, or two different estimands?

One definition, or thirty-four?

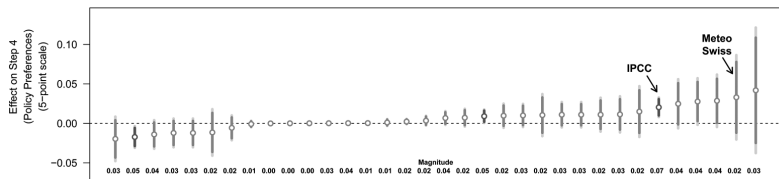
The authors run the same design, on the same people, with the same outcome, under 34 defensible definitions of extreme weather.



Adapted from Quoß & Rudolph (2026), *JOP* 88(3), figure 6. Each dot is the coefficient from one definition, sorted low to high. Bars are 95% and 90% intervals. Filled dots are significant at 5%.

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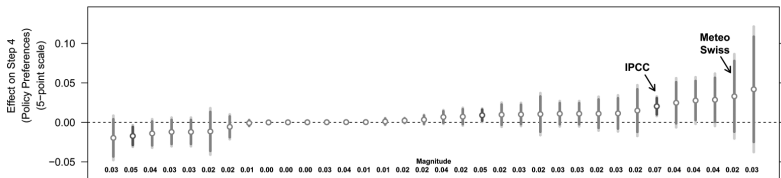


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A single operationalization can support a directional claim. The full set is mostly consistent with a null.

Reducible and irreducible, at one point

Take one fitted curve $\hat{f}_D$ and one value x . Draw a fresh outcome Y at that x and score the prediction.

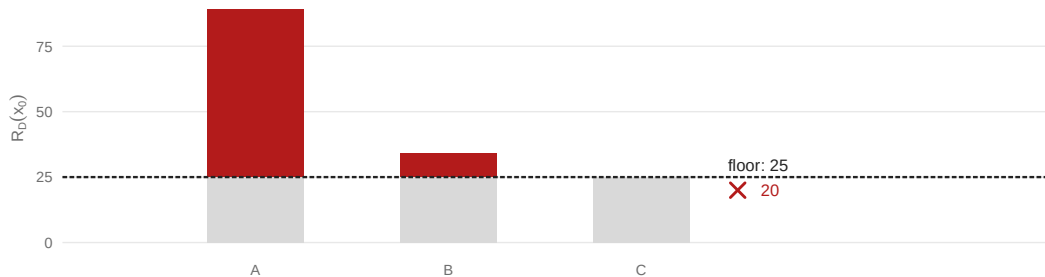
$$\begin{aligned} R_D(x) &= \underbrace{E[(Y - \hat{f}_D(x))^2 \mid X = x, D]}_{\text{average squared miss on a fresh draw, with } D \text{ held fixed}} \\ &= \underbrace{[f(x) - \hat{f}_D(x)]^2}_{\text{reducible: how far this curve sits from the truth}} + \underbrace{\sigma^2}_{\text{irreducible: noise around } f} \end{aligned}$$

At one point x_0 , three fitted curves score $R_D^A(x_0) = 89$ $R_D^B(x_0) = 34$ $R_D^C(x_0) = 25$, and we are told $\hat{f}_C(x_0) = f(x_0)$.

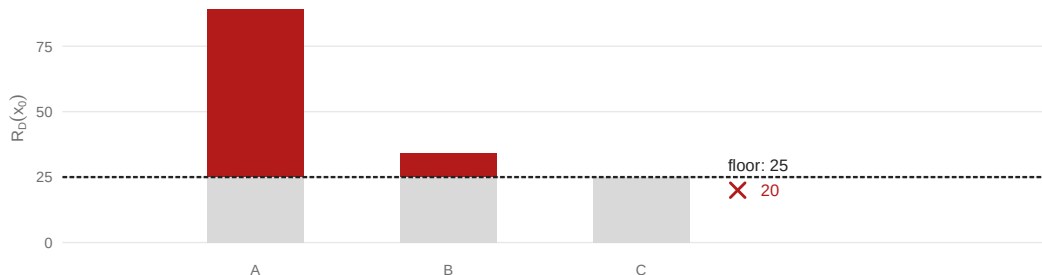
- 1 Find σ^2 .
- 2 Find $|f(x_0) - \hat{f}_D(x_0)|$ for A and B .
- 3 Does curve B sit above or below $f(x_0)$?
- 4 Could a fourth curve score $R_D(x_0) = 20$?

?

Reducible and irreducible, solved

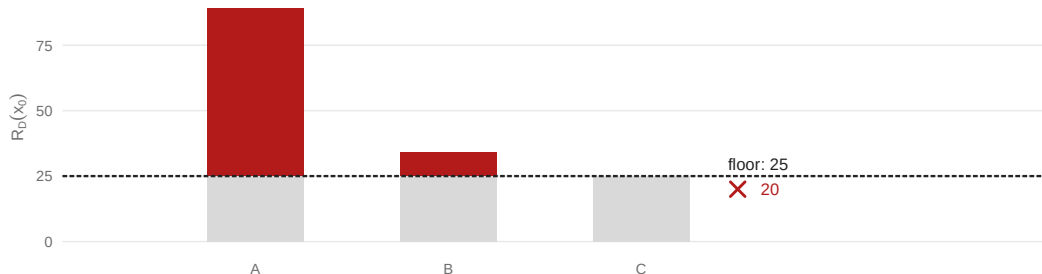


Reducible and irreducible, solved



Curve C has no reducible part, so its whole score is the floor: $\sigma^2 = 25$. Then $\sqrt{89 - 25} = 8$ and $\sqrt{34 - 25} = 3$.

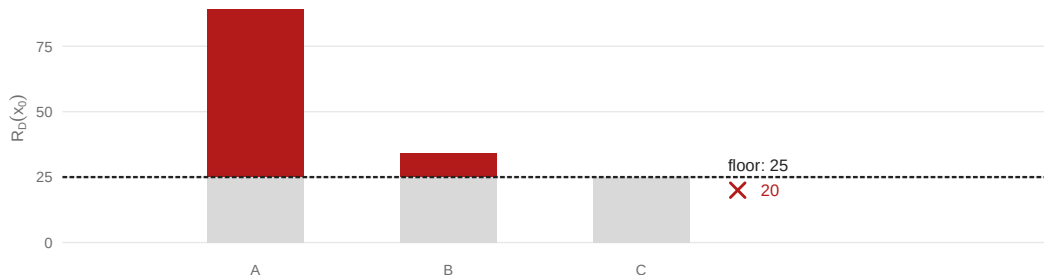
Reducible and irreducible, solved



Curve C has no reducible part, so its whole score is the floor: $\sigma^2 = 25$. Then $\sqrt{89 - 25} = 8$ and $\sqrt{34 - 25} = 3$.

Above or below is **not identified**: the formula holds $[f - \hat{f}]^2$, and squaring removes the sign.

Reducible and irreducible, solved



Curve C has no reducible part, so its whole score is the floor: $\sigma^2 = 25$. Then $\sqrt{89 - 25} = 8$ and $\sqrt{34 - 25} = 3$.

Above or below is **not identified**: the formula holds $[f - \hat{f}]^2$, and squaring removes the sign.

Since $[f(x_0) - \hat{f}_D(x_0)]^2 \geq 0$, every curve scores $R_D(x_0) \geq 25$. A score of 20 cannot be this quantity.

Complete the pipeline

My research question is _____.

$\theta =$ _____

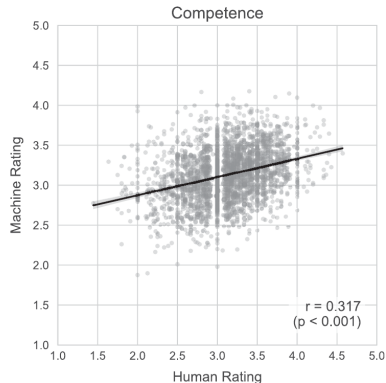
I would estimate it using _____.

The strongest claim this design could support is _____.

If the second line contains $\hat{\beta}$, the name of a regression, or “run OLS”, you have written an estimator rather than an estimand.

Backup

One global error number can hide where the rule fails



$$\mu_i = E_R[\text{rating}_{ir} \mid \text{face}_i]$$

$$\hat{\mu}_i = \hat{f}_D(\text{image}_i)$$

Machine predictions span roughly 2.5 to 3.5 while human ratings span 1.5 to 5.

At which ratings is the rule visibly least accurate?

Adapted from Jiang & Yang (2026), *APSR*, figure 2, competence panel.

A single error number evaluates prediction of human labels. It does not validate the construct, the rater population, or the promotion claim.

Lundberg, Johnson & Stewart (2021), *American Sociological Review* 86(3)

“What Is Your Estimand? Defining the Target Quantity Connects Statistical Evidence to Theory.”

Name the quantity you want before choosing the procedure that will produce a number.

Assigned in this course at lecture 4. Today is the version you can build with slide 9.

$$E[Y \mid A = a] = 10 + 5a + 8(0.5) - 4a(0.5) = 14 + 3a$$

A and Z are independent, so $E[Z \mid A] = E[Z] = 0.5$. Substituting it collapses the z term into the intercept and the az term into the slope.

Used here without proof; Thursday's lecture derives it.

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Quoß, F. & Rudolph, L. (2026). "The Hidden Universe of Operationalizing Natural Experiments: Navigating Extreme Weather." *Journal of Politics* 88(3). Figures 4 and 6.

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Lundberg, I., Johnson, R. & Stewart, B. M. (2021). "What Is Your Estimand?" *American Sociological Review* 86(3).

James, G., Witten, D., Hastie, T. & Tibshirani, R. (2021). *An Introduction to Statistical Learning*, 2nd ed., section 2.1.